

Design History Workbook Module 2 DHF	
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SECTION 1: EXECUTIVE SUMMARY

Acute facial paralysis, more commonly known as Bell's Palsy, is a neurological disorder that causes chronic facial drooping, difficulty closing the eyelid, and dryness of the eye. Currently, the specific cause for the disorder is unknown, however, dysfunction of Cranial Nerve IV and inflammation of the facial nerves have been heavily researched as potential subjects. Bell's Palsy has an annual incidence of 40,000 new cases each year and the lifetime risk is 1 in 60 [1]. Additionally, there is an 8% to 12% recurrence rate [1]. Although this disease isn't ubiquitous, paralysis lowers the quality of life of those affected by making everyday tasks more difficult. For instance, the drooping of the face hinders their ability to communicate effectively with others which greatly diminishes a person's self-esteem and confidence. Additionally, other side effects of Bell's palsy include pain behind the ear, headaches, or difficulty eating or drinking. Eyelid paralysis leads to an inability to blink which can lead to corneal damage and dry eyes which is both painful and affects vision. While many symptoms of Bell's palsy are treatable the loss of function of the eyelid has limited treatment options.

In order to mitigate most adverse effects of Bell's Palsy, common treatment options are corticosteroids, physical therapy, and surgery. Typically, acute facial paralysis is treated using different over-the-counter medications that reduce inflammation and prevent potential infections. The target customers are the demographics with the highest incidence of Bell's Palsy. Typically, it commonly occurs in people aged 40 to 50, pregnant women, and people who have a predisposition to diabetes and respiratory infections also being high risk. Therefore, the specific demographic for Bell's palsy treatments would be individuals between the ages of 15 and 60, especially those in the 40-50 age range and those who have certain medical conditions like diabetes or respiratory infections.

The current solutions for Bell's Palsy aim to improve the symptoms and promote recovery. Antiviral medications can help treat the underlying viral infection that may have caused Bell's palsy, while corticosteroids can help reduce inflammation and swelling in the facial nerve. Physical therapy can help improve facial muscle strength and prevent muscle atrophy. Surgery is also a common treatment option for Bell's palsy. Common surgeries include cross-facial nerve grafting, muscle transfer, and compression surgery. In terms of future directions, there are many new emerging technologies including electrical stimulation, stem cell therapy, and gene therapy. The electrical stimulation technology currently has limitations surrounding the timing of onset and duration of the treatment. Our group is looking to bridge this gap by developing a technology that takes the electrical impulses from the eye not affected by the paralysis which will activate the shutter system closing the other eyelid at the same time. Our solution will bridge the gap of the time of onset, by having both of the eyelids closing at the same time. Similarly, the solution our group came up with is viable long-term in contrast to the existing solutions.

In order to test the effectiveness of our solution in bridging the gap of flaws in the current electrical stimulation technology, we measured the expected blinks as a function of actual blinks. Measuring the blinks required an INA with a notch filter, electrodes placed on the inner eye, an oscilloscope, an Arduino, and a motor with the revolving eye. Essentially, the input signal of the participant blinking was recorded from the electrodes, and this signal was then transferred to both the oscilloscope and the Arduino. If the signal is clean and large enough it would trigger the 3D-printed eyeball on the motor to rotate towards the shutter which simulated a blink. A total of three trials were for each group member with each trial being 30 seconds to a minute. After all of the data was collected, the video of the recordings was played back and the number of actual blinks vs. the number of expected blinks was documented. An ANOVA test was then performed to see if there was a statistical difference between each of the participants as well as between expected blinks and actual blinks. Given our p-value of 0.643, and 0.988 respectively, we can conclude that there was not a statistical difference between each participant which means that subject variables such skin moisture, makeup and eye shape do not

affect the accuracy of our design, thus the model is applicable to anyone.

In terms of future work, there are several areas that can be analyzed, the first being the noisiness of the signal. Adjusting the programming of our solution to automatically detect noise would limit user input and a large source of frustration. Furthermore, the manipulation of a patient eyelid instead of a mock-up would be a major step toward truly providing a device that would solve one of the major issues with Bell's Palsy. Additionally, more research is needed on the optimal electrode placement for effective stimulations, as well as the development of electrodes that are sufficiently sensitive and do not need periodic adjustment.

SECTION 2: PROBLEM STATEMENT AND CLINICAL NEED STATEMENT

2.1 Clinical Need Statement

The primary scope of the clinical need is to correctly identify and treat the condition in order to avoid complications and return face function to normal. Although the exact cause of Bell's palsy is unknown, it is thought to be connected to autoimmune diseases, inflammation, or viral infections. The facial nerve is in charge of managing the facial muscles, including those that manage saliva and tear production, taste, and facial emotions. A damaged or irritated nerve may result in facial weakness or paralysis. Although the precise pathophysiology of Bell's palsy is not fully understood, it is believed to be related to the nerve's swelling and inflammation within the constrained facial canal. Bell's palsy is often diagnosed clinically, depending on the patient's symptoms and physical exam. The signs typically come on abruptly and include facial weakness, drooping of the mouth or eyelids, inability to blink or close one eye, drooling, and trouble expressing oneself facially or verbally.

The current treatments for Bell's palsy included a variety of pharmacological and non-pharmacological interventions. The primary purpose of all these is to simply manage the symptoms and not permanently restore facial function. The most common type of over-the-counter medicines are corticosteroids such as prednisone. These reduce inflammation and swelling of the facial nerve. Antiviral drugs like acyclovir may also be recommended, however, there is still disagreement about how well these work to restore facial function. The use of surgery, face massage, electrical stimulation, and physical therapy are examples of non-pharmacological therapies.

Although the common techniques for treating Bell's palsy have been proven to mitigate the symptoms of paralysis, they have shortcomings in long-term effectiveness, inconsistencies with improvement, long recovery time, and potential adverse health effects. Typically, facial grafting surgeries are two-stage processes that span over the course of 18 months with a 6-month recovery period and a nerve regeneration rate of less than 1mm/day [3]. Nerve grafting is also invasive surgery where a nerve is harvested from the leg and connected to the facial nerves. Due to the extensive recovery period and invasiveness of the procedure, there is a need for the development of new technologies. Similarly, the use of corticosteroids is associated with various side effects, including weight gain, hypertension, and hyperglycemia. A study showed that out of 447 patients, 28% of those allocated to corticosteroids had incomplete recovery of facial motor function after six months [4]. The lack of success and the side effects are reasons why over-the-counter medications are not adequate solutions to the problem. The usefulness of antiviral drugs in treating Bell's palsy is also unknown due to a lack of research, and their usage is not broadly accepted. Therefore, in order to improve the quality of life for those who suffer from Bell's Palsy, it is vital that there is improvement in the longevity, invasiveness, and effectiveness of current treatment options.

2.2 Engineering Problem Statement (Statement of Work)

The current Bell's Palsy treatment techniques are limited in their effectiveness and longevity, resulting in a lower quality of life for those who are affected. An improved Bell's palsy treatment needs to not only mitigate the symptoms of swelling, drooping, and eye dryness, but also improve the functionality of the damaged nerve permanently. The system our group develops must precisely and accurately stimulate facial nerves in order to restore facial muscle function. Critical components of the design that must be accounted for include the proper stimulation to the correct facial nerves, adequate strength, and frequency of the stimulation, and monitor how the facial muscles react to the stimulation. The design will need to perform a number of technical tasks, such as designing an ideal stimulation waveform, an efficient and dependable signal processing system, and a mechanism for controlling the motor that rotates the eye to the closed position. In order to accurately replicate the shutting of the eyelid and determine our improved electrical stimulation techniques' effectiveness, the onset time of the eyelid closing based on the signal from the other eye being closed must be anatomically reasonable. When designing the circuit, it must accurately attenuate the signal so the Arduino can detect the input signal and cause the motor to rotate the 3D-printed eye to the shut position. The model must also yield a high percentage of actual blinks vs expected blinks, be cost-effective, and have long functional capabilities. To assess the number of actual blinks, a recording was taken for one minute per trial per person and the number of times the 3D-printed eye rotated to that position to the correct position signified a blink. After collecting the data, statistical analysis can be done to determine the effectiveness and significance of the experiment.

SECTION 3: PROBLEM DESCRIPTION

3.1 Summary of Clinical Problem

Bell's palsy is a neurological disorder that causes paralysis of the muscles on one side of the face. This can lead to drooping of the face, difficulty with facial expressions, and difficulty eating/drinking, and speaking. A definite cause has yet to be identified, but the condition is linked to inflammation or damage to the facial nerve. While Bell's Palsy can affect any demographic, it is most common in those aged 15 to 60 years old. The normal function of the facial nerve is to control the muscles of the face, allowing for facial expressions, eye closure, and control of the mouth and lips. The facial nerve is responsible for carrying signals between the brain and the muscles of the face, and when this nerve is damaged or inflamed, it can result in the symptoms associated with Bell's palsy. The pathophysiology of Bell's palsy involves inflammation or damage to the facial nerve, which can be caused by a variety of factors, including viral infections, trauma, or autoimmune conditions. This damage can lead to decreased blood flow to the nerve, causing swelling and compression, as well as damage to the myelin sheath that surrounds the nerve, leading to further inflammation and loss of function. In terms of the clinical presentation, Bell's Palsy usually has a sudden onset of paralysis on one side of the face which progressively worsens with time. Other symptoms can range from the drooping of the eyelid and mouth, the inability to blink or close the eyelid, loss of taste, and difficulty hearing.

Typically, these symptoms are mild, however depending on the severity of the paralysis, these symptoms can worsen. Similarly, these symptoms can cause emotional and social stress due to the prominent face changes and lack of ability to communicate. When looking at the clinical outcomes of Bell's palsy they vary on the patient-by-patient basis. Some patients can make a full recovery in several weeks while some can take several months. In more severe cases, some patients experience long-term paralysis of the affected muscles, as well as difficulty with speech and eating. Based on the number of

incidences annually Bell's Palsy is relatively common. Currently, approximately 1 in 5,000 people each year in the United States is affected by this [1]. As mentioned previously the exact cause of Bell's palsy is not well understood, it is believed to be related to viral infections. When looking at the economic impact of Bell's palsy the first thing that must be considered is the loss of productivity. This condition often requires time off from work and school, in addition to the cost of medical treatment and rehabilitation. According to a study published in the journal BMC Neurology, the direct medical costs of treating Bell's palsy in the United States were estimated to be over \$300 million per year, with indirect costs such as lost productivity and disability accounting for an additional \$1.5 billion per year [5].

3.2 Summary of Current/Emerging Solution Landscape (1/2 pg)

Succinctly summarize the existing solutions along with their risks and benefits. This summary could be in the format of text, a table, and/or a figure. Be sure to document your search with proper citations. While the current solutions for Bell's palsy, such as corticosteroids, antiviral medications, nerve grafting, and physical therapy, have been shown to improve outcomes in some cases, there are several shortcomings that highlight the need for an improved electrical stimulation solution. The main issues with the current treatment options are the inconsistent effectiveness and long-term feasibility. For instance, while corticosteroids have been shown to improve outcomes in some patients, their effectiveness is limited to specific demographics. Similarly, antiviral medications have been shown to have a similar trend. Nerve grafting, another potential solution, is a surgical procedure that involves transplanting healthy nerves from other parts of the body to the affected area of the face. While this approach can be effective in some cases, it is an invasive procedure that carries significant risks and is not suitable for all patients. Physical therapy is another potential solution for Bell's palsy, which involves exercises designed to improve muscle strength and coordination in the affected area of the face. While this can work in some patients this isn't a long term viable option due to the time commitment. Overall, the current solutions for Bell's palsy have significant limitations. This highlights the need for an improved electrical stimulation solution that can provide more consistent and effective outcomes. An improved electrical stimulation solution could potentially provide a less invasive and more targeted approach to treating Bell's palsy, specifically eyelid paralysis, with fewer side effects and a higher success rate than current treatments.

	Description	Pros	Cons
(Base Line) Corticosteroids	Corticosteroids are a class of drugs that reduce inflammation and swelling, and are commonly used to treat Bell's Palsy. [6]	• Effective in reducing inflammation and swelling • can improve recovery time and symptom relief [6]	• May cause side effects such as increased blood pressure • weight gain • mood changes • not effective for all cases of Bell's Palsy [6]

Nerve Grafting	Nerve grafting is a surgical treatment that includes the attachment of a healthy nerve from another section of the body to the injured face nerve. The surgery is often used in severe instances of Bell's Palsy that have failed to respond to conventional therapies. [7]	• Can provide long-term relief of symptoms • improves facial muscle control and symmetry [7]	• Invasive surgery • requires an experienced surgeon • potential nerve damage • infections and scarring • [7]
Electrical Stimulation	Electrical stimulation uses a low-level electrical current to stimulate the muscles in the face, helping to improve muscle tone and reduce the appearance of facial drooping. [8]	• Non-invasive • can be done in outpatient settings • can be used to target specific muscles • provides relief of symptoms and may improve recovery time. [8]	• Results vary between individuals • requires specialized equipment • may not be covered by insurance. • Can irritate the facial muscles [8]
Antiviral medications	Antiviral medications are used to treat viral infections, including herpes simplex virus, which has been linked to Bell's Palsy. [9]	• Can help reduce the severity and duration of symptoms, • may be effective in treating underlying viral infections. • Covered under insurance [9]	• Must be taken early in the course of the illness • may not be effective for all cases of Bell's Palsy • side effects may include nausea and headache [9]
Physical therapy	Physical therapy for Bell's Palsy typically involves exercises and stretches that target the facial muscles and improve muscle tone and function. [10]	• Non-invasive • can help improve muscle control and symmetry • may improve recovery time and relieve symptoms [10]	• Results vary between individuals • requires consistent effort and commitment to exercises • may not be covered by insurance [10]

SECTION 4: DESIGN SPECIFICATIONS

Acute facial paralysis is a neurological disorder that causes a myriad of symptoms, but this solution will be focusing on partial eyelid dysfunction. The solution will utilize the electrical impulses from the functioning eyelid to stimulate the eyelid affected by Bell's Palsy. This will allow the patient to blink and close their eyelids properly which will prevent the pain, infections, and required treatments associated with dry eyes. The device would be implanted and continuously active, so it would meet the users needs in terms of when and where. Current treatment methods include steroids, antiviral medicine, analgesics, physical therapy, and eye drops, but none attempt to directly correct the physiological movement of the eye.

4.1 Target Customer and Rationale

With this in mind, the major stakeholders in this design would be those who are affected by the disease, surrounding friends and family, primary care physicians, nurses, neurologists, physical therapists, scientists, government officials, hospitals, insurance companies, businesses, etc. These people would play a role in the development of the device in terms of design, testing, approval, regulation, distribution, selling, usage, and various other aspects. The main populations under consideration when setting the design specifications are the customers and developers. Patients would benefit from improved eye function and medical groups and professionals would need to learn and adopt the new method to provide better options that could improve personal, social, and economic states. Hypothetically, improved electrical receive and stimulation technologies can also be applied to a wide range of needs for populations with muscle and nerve dysfunction.

The group decided to test and analyze different techniques for signal retrieval, filtering, and amplification from the functioning eyelid as well as how to stimulate the other eyelid using specific programs.

4.2 Summary Table and Discussion of Design Specifications

Design Criteria (Customer or societal need)	Design Specification
Must use electrical input from eyelid	The electrical input from the eyelid was collected using electrodes placed on the top and bottom eyelids. The mV range of stimulus differed by person and impedance. When the electrodes were placed, the peak values were determined when the person blinked and that was used to signify stimulation of the other eyelid. Instrumentation amplifiers (1000 gain) and filters (notch 60Hz) were used to make the input signal easier to utilize.
Must stimulate other eyelid consistently/repeatedly	A higher voltage was seen when the person would blink which would cause the Arduino to react and signal using a light. This showed proof of concept, and it was implemented to turn a motor and model actual blinking with a physical prototype. Data was taken to compare actual blink counts to the devices simulated blink counts across multiple repeats and replicates. The goal was to have no significant difference in error across all samples and across subjects.

Must match speed of eyelid movement accurately	The speed of the simulated eyelid movement needs to simulate the speed of actual eyelid movement. These values would be hard to measure in a lab of this caliber, but the motor and gear mechanism used to model the blink was designed with respect to the typical 45mm/s for a blink.
Must match timing of eyelid movement accurately	The timing of the eyelid corresponds with the delay in simulated blink compared to the original stimulus. This was a stretch goal for this design, and count was used in place of time. We would like our model to accurate roughly 80% of the time
Must be easy to use	The device has to be easy to implement. It takes precision to set up originally for the thresholds and placement, but once put in it should be easy to maintain and ideally no alterations would need to be made. Ideally setup takes under 10 minutes.
Must be cost-effective	The cost of the design has to be low compared to alternative methods of treatment. It would be composed of an electrode to observe the original signal, a method to send it to the other eye, and an electrode to stimulate the paired eyelid. It would be implemented once and left in for an extended period of time, so as to save from repeated costs. Cost should be under 1000\$ per year based on current treatment.

When looking at the design criteria and specifications, the most important aspects in this design laboratory are that it must use the electrical input signal from the base eyelid and that it must effectively stimulate the other eyelid. Being able to read the signal sent and signifying a response proves that it is functioning properly, while the other aspects are used to enhance its function.

SECTION 5: SOLUTION STATEMENT & EXPERIMENTAL DESIGN

5.1 Overall Solution Statement

Given our design problem, our group decided that our solution would center around the detection, purification, and stimulation of an electrical signal from electrodes placed near the eye. These electrodes would capture the signal that corresponds to activation of the orbicularis oculi muscle, which is responsible for the blinking motion of the eye, specifically the closing. This would then be improved using an instrumentation amplifier and filter to clarify the signal. Then it would be analyzed and each activation point or peak in the signal would correspond to a mechanical movement in the mock-up eye system. This represents the electrical stimulation of an actual human eye. The system would utilize the signal from the healthy eyelid and send it to the paralyzed eyelid causing it to effectively blink.

The first round of iterations took place during weeks one, two, and three and revolved around the development of the circuit. The group knew that it needed to amplify the signal using an instrumentation amplifier, but needed to determine what gain was necessary and how it needed to be split up. Originally, a gain of 1000x was determined to be the most ideal and we considered splitting it into two stages but settled on just one. Multiple resistor values were tested to create the proper gain as the exact resistors were not available in the lab. Various tests were run with different inputs until the 1000x (closer to 1.3k) gain was achieved. When this was actually implemented into the final circuit, the gain was closer to 100x for an unknown reason, but it still allowed for the circuit to function properly so

it was kept. The group knew that the signal needed to be filtered in order to be utilized properly, so testing was conducted on both notch and low pass filters. Early on, the low pass filter was taken out as it was not necessary to have a proper signal read. The notch filter was iterated multiple times with different resistor and capacitor values. In an ideal situation, it would filter around 60Hz, but when implemented it was closer to 64Hz. With multiple attempts, this was lowered to 62Hz and then finally about 60.5Hz. The INA and notch filter that was used can be seen in the figure below. When these were tested with both a simulated input as well as an actual blink, the output showed the appropriate stimulus.

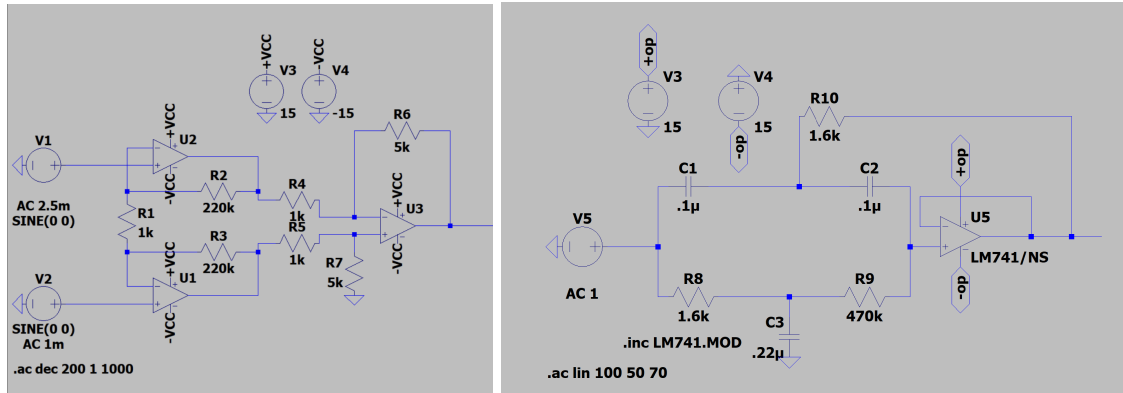


Figure 1. INA and notch filter.

The eye and eyelid systems were iterated during weeks two, three, and four. These sections were more straightforward in terms of completion because they were focused on showing proof of concept rather than an implantable device. The base of the eye was modeled in three parts to ensure symmetry and proper production, but when it was actually printed it came out effectively in one piece for the eye and one for the box to hold it. The eyepiece was kept, but the box was discarded because of iterations to the eyelid mechanism. The eyelid mechanism underwent multiple iterations. The first design resembled a blind system that was attached to a string to lower over the eye and then retract when signaled. This design was too difficult to implement, so an alternate solution was developed. A gear system replaced the old blind system and a piece of paper and tape was attached to the extruding piece which symbolized the eyelid. When attached to the motor via a washer and screw, the gear rotated correctly around the eye and simulated a blink.

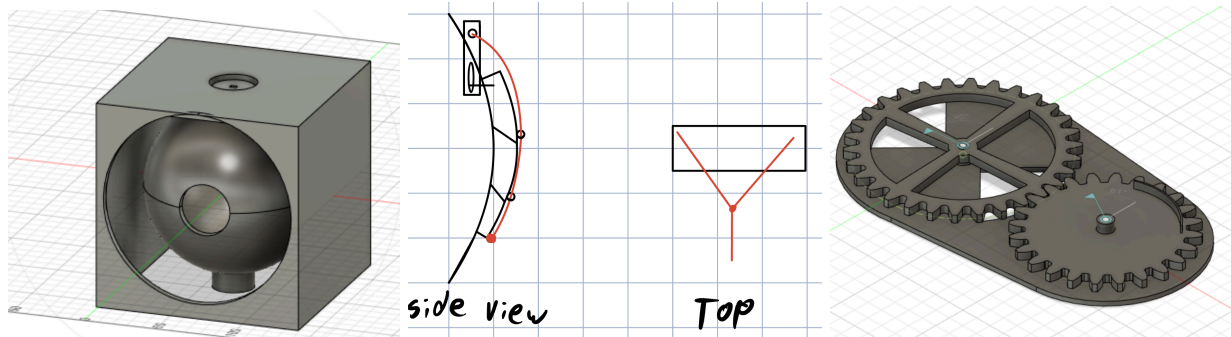


Figure 2-4. Eye and eyelid models.

The code for the Arduino took the longest amount of time to iterate, as it combined the detection of the input signal, the output signal and stimulation of the light, and then the movement of the motor to simulate a blink. This occurred during weeks three, four, and five. The code took the input signal of the eye and when the participant blinked, it sent a peak signal. This difference in voltage signified a blink to the software, which underwent different commands to signal a light to show the user that it was detected. Further in development, this light was coupled with the turn of a motor using the apparatus above to show the physical and mechanical movement of a modeled mock-up eye blinking.

```

1 // int forward = 2;
2 // int reverse = 1;
3
4 bool ignore = false;
5 bool detect = false;
6
7 unsigned long firstPulse = 0;
8 unsigned long secondPulse = 0;
9 unsigned long interval = 0;
10
11 const int steps = 20;
12
13 Stepper motor = Stepper(steps, 8, 9, 10, 11);
14
15 int sensor = analogIn(A0);
16 float voltage = sensor;
17
18 void setup() {
19   // put your setup code here, to run once:
20   Serial.begin(9600);
21   pinMode(7, OUTPUT);
22 }
23
24 void loop() {
25   // put your main code here, to run repeatedly:
26   if (reading > 0) {
27     digitalWrite(7, HIGH);
28   }
29   //waiting edge
30   if (reading > upper limit ignore == false) {
31     if (detect == false) {
32       firstPulse = millis();
33       detect = true;
34       digitalWrite(7, HIGH);
35       //first pulse delay (100ms)
36       //move motor down
37       //motor-step(steps);
38       //delay(1000);
39     }
40     //move motor up
41     //motor-step(steps);
42     //delay(1000);
43     ignore = true;
44   }
45   if (reading < lower limit ignore == true) {
46     ignore = false;
47     Serial.println(voltage);
48     digitalWrite(7, LOW);
49   }
50   Serial.println(0);
51   Serial.println(voltage);
52 }

```

Figure 5. Arduino code.

5.2 Functional Block/Subcomponent Design

Subcomponent Title	Technical Description	Individual Responsible	Design Specification Addressed
Eye system	3D modeling and printing of the eye base that the moving components will be anchored to. Multiple prints will be done if there are any spherical malformations or inconsistencies.	AH	Simulates ease of use and eyelid stimulation
(Eyelid) gear system	3D modeling and printing of the eyelid gear system that comprises the moving components. Connection points to the motor and base will be evaluated for proper linkages. Interlocking mechanism and rotation will be tested to ensure efficacy.	KR	Simulates ease of use and eyelid stimulation Matches eyelid speed and timing
Circuit	Testing of proper gain and filtering will be conducted with	RS, KR, AH	Uses electrical signal input and output

	a waveform generator to prove the concept. Then it will be tested with human signal input for each group member to make sure that it works properly and provides the proper output signal on the oscilloscope.		Must be cost-effective
Arduino	Input signal and proper difference of voltage will be verified with the blinking of a light and then the trigger of the motor.	AH	Uses electrical signal input and output Matches eyelid speed and timing
Motor	Motor movement will be evaluated based on degree of rotation, direction, and consistency.		Matches eyelid speed and timing

The eye model is attached to the center of the gear of the eyelid mechanism so that the extruded piece shifts over it like a simulated eyelid. The gear is attached to the motor using a washer and screw. The motor is attached to the Arduino board which is connected to the computer with the appropriate code. The computer is connected to the circuit which sends it the proper signal that is retrieved as the input from the participant. This is done via electrodes on the top and bottom eyelids. This describes the interactions between each subcomponent.

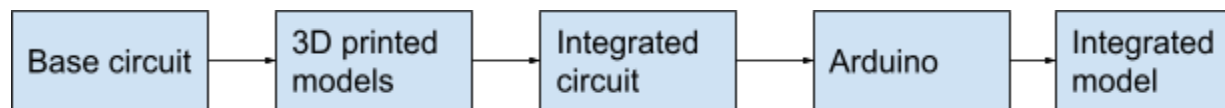


Figure 6. Subcomponent flowchart.

The base circuit was generated alongside the design of the 3D printed models. The circuit was then tested with simulated and biological inputs to ensure efficacy. This was then attached to the Arduino to produce the correct output. This signal was then sent to the integrated model so that it could stimulate a light and motor to simulate a mock-up eye blink. This process can be seen in the flowchart above.

SECTION 6: FAILURE MODES AND EFFECTS ANALYSIS (FMEA)

The FMEA table below displays failure modes for each subcomponent that could either affect our prototype results or pose a potential risk of injury to ourselves. Risks of injury failure modes have higher severity scores due to the potential of an electrical shock around the eye that could cause long-term injury. This is why a short circuit is one of our most severe failure modes. Since the cause for this is mostly human error the only way to mitigate it is through procedures. Our biggest risk failure mode risk is an incorrect circuit. The FMEA analysis influenced our decision to change the design of our notch filter. After running a frequency sweep we noticed that our cutoff frequency wasn't close enough to our desired cutoff frequency so we tweaked the resistor values to get our cutoff frequency closer to 60Hz. Our third biggest potential failure is the bad placement of electrodes. Bad placement is a potential failure mode because the cause is the anatomical structure of the subject. This means that one electrode

position on one subject won't be functional on another subject. The only way to fix electrode placement is by trial and error because anatomy and skin moisture can all affect conductivity and the viability of the results. Another problem we had with the electrodes was that the electrodes that are in the lab are too large and don't have strong enough adhesive to stay on the subject's face. Based on this failure mode we decided to order electrodes meant for facial applications.

The FMEA table produces a risk priority number with three components. Each scored on a scale from 1 to 10. The first component is severity, where one is the least severe and ten is the most severe. The second is the occurrence likelihood, where one is least likely to occur and ten is most likely to occur. The last is detectability, with one likely occurring and ten least likely.

Process/Step/ Input/Output	Potential Failure Mode	Potential Failure Effect	S	Potential Causes	O	Current Controls	D	RP N	Recommended Mitigation
circuit	incorrect build	inaccurate results	6	human error when building	6	LTSpice	6	216	Using a protoboard instead of a breadboard
	short circuit	damage to components	7	Human error when building	5	LTSpice	6	210	following a procedure when connecting power supply
		risk of injury	8	Human error when building	3	not connecting electrodes when shorting	6	96	following procedure to fixing a short
	bad connection to electrodes	risk of injury	5	Human error when connecting equipment	2	changing out electrodes	3	30	find two electrodes that work and only use those
		inaccurate results	5	human error when connecting equipment	5	changing our electrodes	1	25	checking connections and changing out back cords
	incorrect solder	inaccurate results	6	human error when soldering	1	following PCB diagram	5	30	build circuit before soldering
		risk of injury	6	human error when soldering	1	skill when soldering	3	18	clear, workspace, and correct PPE
Arduino code	Incorrect noise cutoff	inaccurate results	7	skin condition of subject	5	trial and error testing different noise cutoffs	1	35	modify arduino code to find accurate noise cut off
	not uploading changes to circuit	changes in the code won't change the output	3	human error	2	uploading immediately after making changes	1	6	doubling checking upload was successful before beginning trial
electrode placement	bad placement	unable detect blink	4	anatomy of the eye	6	trial and error when placing electrodes	2	48	using electrode gel to increase conductivity

		uncomfortable placement	2	anatomy of the eye	7	repositioning electrodes	1	14	following procedure based on anatomy
	electrodes unable to stay on	unable to obtain signal	3	adhesion not strong enough	8	changing electrode patches	1	24	using electrodes that are meant for facial applications
Motor and gear system	Screw is too loose between gear system and motor	system fails to spin	4	erosion of model due to the screw and motor	2	adjusting screw until the system spins	1	8	repringing gear base and motor gear to mesh better
	Burning out the motor	broken motor	5	applying too much torque on the motor	1	switching out the motor	3	15	setting the RPM low enough to not burn out the motor

S =Severity of failure; O=Occurrence likelihood; D=Detectability; RPN=Risk Priority Number=S×O×D

SECTION 7: VERIFICATION AND VALIDATION OF DESIGN

7.1 Design Verification Plans

We used three procedures to verify our solution's function and one to validate that our solution will meet consumer needs. We verified our design choices by using LTSpice to ensure we met are design criteria before building our circuit. We used LTSpice to specifically verify that our INA has a gain of 1000 and our notch filter attenuates at 60Hz by running a peak-to-peak analysis of the voltage and then calculating the gain by dividing the output peak-peak-value by the input-peak-to-peak value for the INA. The Notch filtered was verified by running a frequency response and plotting the bode graph. The INA is necessary for amplifying the output signal of the eye so that it is readable to the arduino and the notch filter attenuates the 60Hz noise from the power supply, cleaning up the signal. Figures7-8 show the equation for the mathematical model and the circuit diagrams

$$A_v = \frac{V_o}{V_2 - V_1} = (1 + \frac{2R_2}{R_{gain}})(\frac{R_4}{R_3})$$

Figure 7: Equation used for INA calculation

$$f_H = \frac{1}{2\pi R_2 C_2} \quad f_C = \frac{1}{2\pi RC}$$

Figure 8: Notch filter equations

Our second verification procedure was verifying that each electronic component was within their respective tolerances. This ensures that if there is a failure in the circuit it is not due to faulty electronic parts. It also ensures the accuracy in obtaining design specifications. Each component was measured with a multimeter and then the measured value was evaluated by finding the percent error.

Third, we validated that our breadboard was producing the results from LTSPICE. Analyzing the functionality of our circuit ensures that the circuit works as expected and allows us to acquire accurate data when testing. However when verifying the functionality of the INA we found that our circuit was not amplifying as expected which can be seen in table 3, but when testing it with our eyes we found that the blink was still visible on the oscilloscope. while the INA verification does not match the mathematical model it still meets design specifications. Verifying our notch filter went as expected as seen in Figure _. The notch filter attenuates the 60Hz house from the power supply, thus making the signal cleaner.

The accuracy of our entire solution, including the circuit, program, and mechanical system also required verification. To test this, we used a function generator to output a 10 mV pulse wave at 60 HZ, which was input into the circuit. The circuit design was the same as Figures 1 above, while a screenshot of the code is shown in Appendix 9.3 "Code". The code is a relatively simple peak detection algorithm, stipulating that every single time it is run, the cut-off frequency needs to be changed. This code was loaded into an Arduino, either a UNO Rev3 Wifi or Rev2 were used. When connected together, the circuit would amplify the signal from the eyelid,

After we verified that our solution, when presented with ideal conditions, performed with 0% error, we then validated our solution by connecting electrodes to each of the group members. Our group members were composed of 2 men and 2 women, with varying levels of skin-care routines, and took place across different days, but around the same time of day. The number of factors did not end here, with contact lens use and hat usage also recorded, but the result of this combination of factors is that we could validate the accuracy of our design solution across most of the conditions that we would expect potential patients to be in.

The error would be calculated by counting the number of actual blink movements made by the design, and the number of blinks that the group member actually did, which we labeled as the number of expected blinks. These results would be normalized by calculating the number of actual and expected blinks per minute per person, which would then be used to calculate the percent error of each group member. Each group member was expected to run 3 trials, each lasting around 30 seconds. The percent error of each trial was calculated using the formula, $\%error = \frac{actual - expected}{expected}$.

To measure the applicability of our design and the accuracy, two One-Way Analysis Of Variance (ANOVA) tests were performed. One was between the percent error between subjects and the other was done between expected blinks and actual blinks. This was done using the Analysis ToolPak Add-In in Excel, and the results of this test would show if there is any difference. Our alpha-value was 0.05, and the null hypothesis that we tested is that there is no statistically significant difference between the % error of the trials. Conversely, the alternate hypothesis is that there is a statistically significant difference between the % error of the trials. For our testing, we are expecting to accept the null hypothesis to show that our design is effective no matter the subject and that more design accurately blinks when it is supposed to.

7.2 Design Verification Testing Results

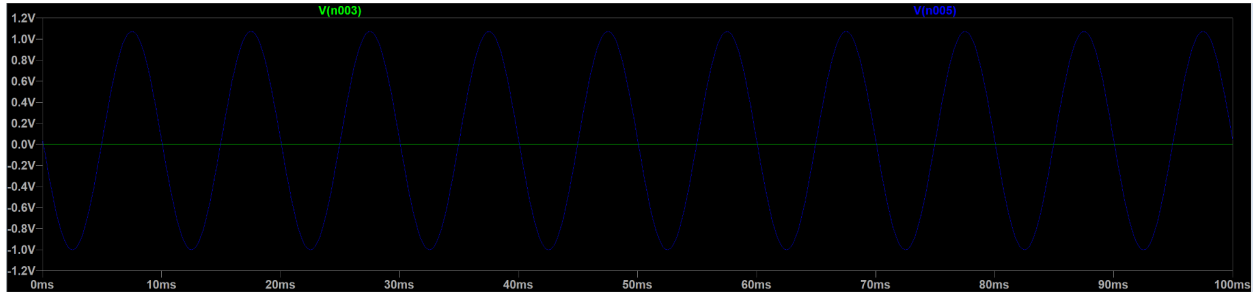


Figure 9: LTSpice Simulation of INA

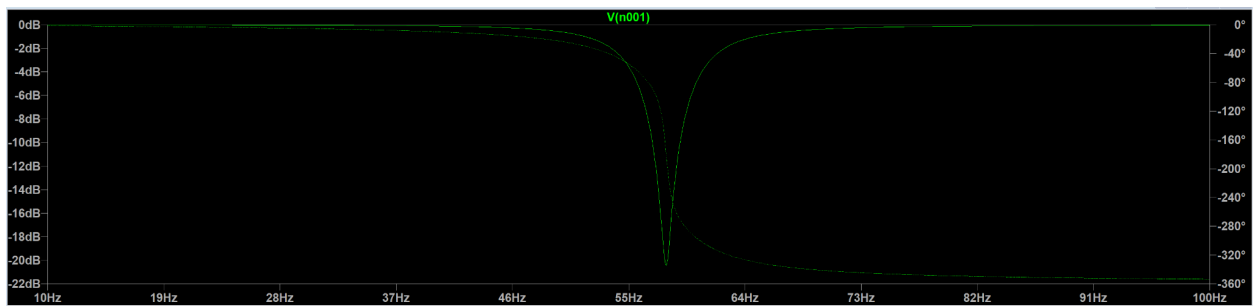


Figure 10: LTSpice Simulation of Notch Filter

Results from LTSpice simulation show that our INA should have a gain of 1.1k and the notch filter attenuates at 58Hz. The INA has a 10% error from our design specification and the notch filter has a 3.3% error from our design specification. both were deemed to be in an acceptable range that satisfies our requirements.

Table 1: INA resistor values

Component	Percent error
R1	1.80%
R2	0.14%
R3	0.05%
R4	1.5%
R5	1.2%
R6	0.17%
R7	0.08%

Table 2: Notch filter resistor and capacitor values

component	percent error
R1	1.5%
R2	0.2%
R3	1.6%
C1	3%
C2	2%
C3	9.5%

Each resistor has under 5% error meaning they are within their tolerance of $\pm 5\%$ and capacitors are within 20% tolerance so all components are deemed acceptable

Table 3: breadboard INA gain verification

Circuit	In	out	Gain
beadboard	47.5mV	8.4V	176.84
LTSpice	1mV	1.1V	1100
		Percent error	84%

The INA does not meet our design specification of 100 gain but it still amplified the signal enough to make the blink visible. The purpose of the INA was to amplify the input signal so that the blink can be recorded and even with less amplification the blink was still visible as seen in Figure 11 thus the INA was

deemed acceptable. The notch filter attenuates at 63Hz which has a 5% error and thus was deemed acceptable.

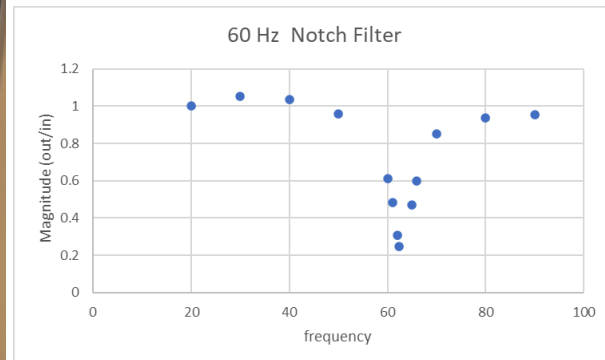
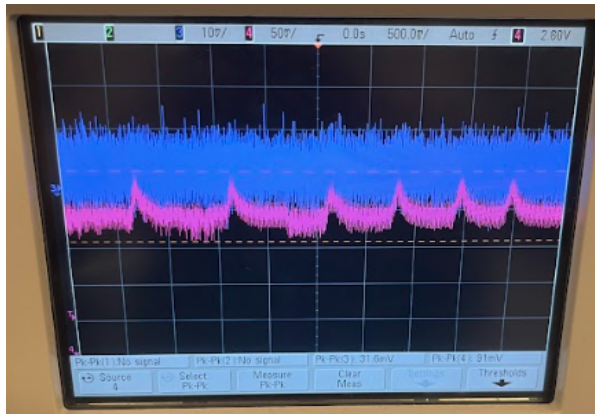


Figure 11-12: verification of breadboard INA and notch filter

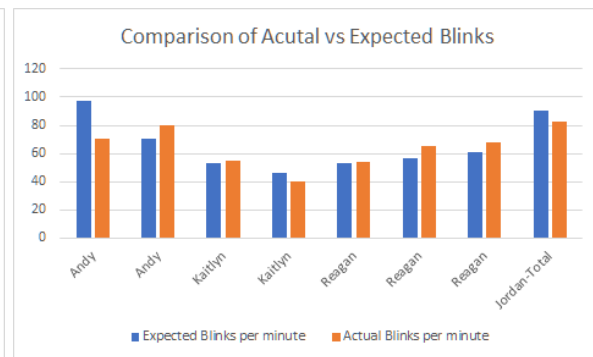
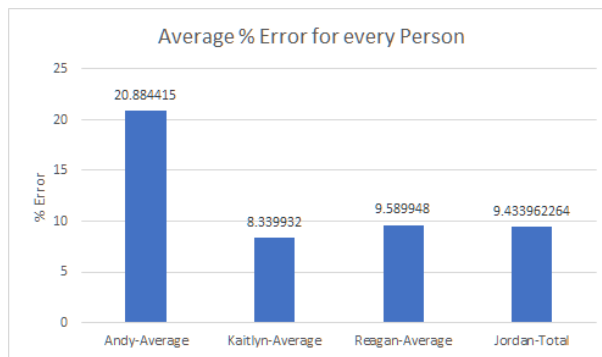


Figure 13-14: percent error and accuracy of results

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
Column 1	2	-14.3716	-7.18579	872.3176		
Column 2	2	-9.40714	-4.70357	139.1089		
Column 3	3	28.76984	9.589948	51.3485		
Column 4	1	-9.43396	-9.43396	#DIV/0!		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	509.9517	3	169.9839	0.610287	0.643	6.591382
Within Groups	1114.124	4	278.5309			
Total	1624.075	7				

Figure 15: ANOVA showing no significance between subjects

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
Column 1	14	561	40.07143	643.3022		
Column 2	14	559	39.92857	774.9945		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	0.142857	1	0.142857	0.000201	0.988784	4.225201
Within Groups	18437.86	26	709.1484			
Total	18438	27				

Figure 16: ANOVA showing no significance between expected blinks and actual blinks

After calculating the percent error for each patient and running the ANOVA tests show with p-values of 0.643 and 0.988 respectively, show that there is no statistical significance between subjects or between expected and actual blinks.

7.3 Discussion Relating Final Design Results to Literature, Design Specifications, and Customer Needs

With respect to our engineering problem, designing a functional eyelid prosthesis, our group was able to show off a practical demonstration of the signal processing and analysis requirements for such a prosthesis. Overall, we found that an INA with a gain of 1000 and a notch filter set to 60 HZ was sufficient to generate a clear electrical signal from electrodes placed on top of and below a patient's top eyelid. This could then be analyzed to find peaks that signified a "blink", which could then be interpreted using Pulse Width Modulation(PWM) to drive a motor modeling a blinking motion. We showed that our model, with 7 trials across 4 patients, resulted in an average %error of 12.06%, with a maximum of 20.88% and a minimum of 8.33%. This is very acceptable and means that for the normal spontaneous blink rate of a person, we would expect anywhere from 1 to 3 blinks per minute to be misidentified. Furthermore, ANOVA testing showed that there were no significant differences between our test subjects, showing that our solution works for most, if not all, skin and anatomical types. As a result, our proof of concept shows that an accurate and widely available eyeball prosthesis could be created.

While we believe these results to be satisfactory, further development and testing need to be done. One major area that needs to be looked at is the code, as currently there is a limit on how fast the code allows a blink to occur. This is a limitation of the programming skills of our group, as we were unable to analyze the input signal to the Arduino and output the PWM at the same time. In addition to this, prior to every new trial, the input signal to the Arduino needed to be analyzed by a group member to find a cutoff value, as the noise in the signal was different every single time. An additional function of the program could be added to where it calculates the average noise every half of a second and automatically sets the cutoff value.

Moving on from programming optimizations, another method of connecting an Arduino to a computer must be found, as the electrical noise from the laptop was being transmitted to the Arduino and messing with the signal. The cause of the noise was not found, and it was not reproducible, but this is still a factor that definitely needs to be addressed. A future direction could be to use different low-power microprocessors such as the commonly available ATmega32u4, which would decrease both monetary and power costs. Finally, instead of a mockup of an eye, future research should be done into manipulating an actual eyelid on the subject. This would result in what our group would consider an acceptable medical device that would truly solve our problem statement, and significantly improve the quality of life of Bell's Palsy patients.

SECTION 8: CONCLUSION

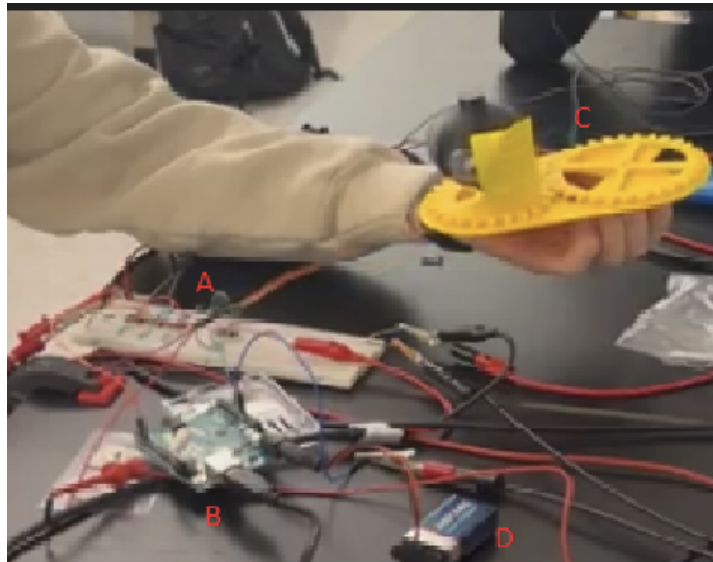


Figure 17: Final Design Solution

As a result of experimentation, iteration, and design changes, the original design has been refined in the pursuit of fulfilling the solution statement. Figure __ above shows the final design solution. Overall, an INA circuit connected to a Notch Filter set to 60Hz should be connected to two electrodes placed near the patient's eye, one right above the cheekbone and one below the eyebrows just touching the upper eyelid. This is shown as A in the photo, with the INA leaning towards the left. The output of the notch filter will be connected to an Arduino running a program that detects the electrical impulse that signals the beginning of a blink, and calibration will be done at this point, as the impedance of each patient's skin is different. The Arduino is labeled as B in Figure 17, with the power supply being a 9V battery labeled as D. The output from this program will then be connected to a mechanical system that simulates the blink of an eye, allowing for validation and verification, which is labeled as C.

Modifications to the initial design solution are numerous but did not change the overall design of the solution. For example, the code was changed over 200 times over the course of testing, with 3 different motors being used since the first two burned out from overcurrent. In addition, the initial INA was soldered onto a protoboard but was damaged beyond repair due to the storage box being dropped. Therefore, a second INA was built on a breadboard to the same specifications, with the Notch Filter also built on the same breadboard. 2 different USB-A to USB-B cables were used because the first one appeared to have a broken shield ground leading to excess noise. Power supplies were also changed to the Arduino, with a 9V providing power instead of the 5V USB standard, which helped to reduce noise. These are some of the minor modifications that were made, but the overall solution design, electrodes connected to an INA, then a notch, then an Arduino running code, and finally a motor remained the same.

The data and analysis of our data show that our solution shows great potential. Over four different trials, the highest level of error was 20.88%, with an average error of 12.06%, which we believe to be quite good. Since these trials were done on both male and female subjects of different ethnicities and different skincare routines, with contact lenses in and without contact lenses, we believe that our solution is adaptable and accurate. With commercially available parts and materials, our solution is also relatively affordable, and we project that with the increase in semiconductors, the price of Arduinos and other microprocessors will only become more affordable. With more time and resources, our group is confident that the manipulation of an eyelid, our next step, could have been completed.

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Appendix A:

PROJECT PLANNING

9.1 Project Schedule

Task	Week 1 (3/7 - 3/13)							Week 2 (3/14 - 3/20)							Week 3 (3/21 - 3/27)						
DHF (Sections 1- 9)																					
Design Presentation																					
Design INA circuit in Lt. Spice																					
Design Notch Filter in Lt. Spice																					
Calculate Resistor Values																					
Calculate Capacitor Values																					
Create CAD of eyeball																					
Perform LT Spice Simulations																					
Construct the Circuit																					
Obtain Signals from eye on Oscilloscope																					
Test Arduino Code																					
Perform Gear System Calculations																					
Test Notch Filter																					
Create Gear System in CAD																					
Test Combined Filter																					
Build Combined Filter																					
Construct Gear System with eyeball																					
Record the trials																					

Task	Week 4 (4/4 - 4/10)							Week 5 (4/11 - 4/17)						
DHF (Sections 1- 9)														
Design Presentation														
Design INA circuit in Lt. Spice														
Design Notch Filter in Lt. Spice														
Calculate Resistor Values														
Calculate Capacitor Values														
Create CAD of eyeball														
Perform LT Spice Simulations														
Construct the Circuit														
Obtain Signals from eye on Oscilloscope														
Test Arduino Code														
Perform Gear System Calculations														
Test Notch Filter														
Create Gear System in CAD														
Test Combined Filter														
Build Combined Filter														
Construct Gear System with eyeball														
Record the trials														

9.2 Project Budget

ITEM	SUPPLIER	PRICE PER UNIT	COST TO PURDUE	ACTUAL COST
disposable cloth facial electrodes	biopac systems	\$0.8 for 1pc	\$48 for 60pk	\$48 for 60pk
adhesive electrodes	3M	\$0.19 for 1pc	\$199 for 1000pk	\$199 for 1000pk
breadboard	electronix express	\$4.75 for 1pc	\$42.5 for 10pk	\$42.5 for 10pk
wiring kit	RSR electronics	\$20 for 1kit	\$20 for 1kit	\$20 for 1kit
cable kit	science purchase	\$18 for 1kit	\$18 for 1kit	\$18 for 1kit

oscilloscope	valuetronics	\$2995	N/A	\$2995
waveform generator	valuetronics	\$1050	N/A	\$1050
multimeter	fluke	\$470	N/A	\$470
power supply	bell electronics	\$795	N/A	\$795
motor	flash hobby/amazon	\$6.3 for 1pc	\$12.6 for 2pc	\$12.6 for 2pc
3D printing materials (PLA)	prusa research	\$25 for 1kg	\$50 for 2kg	\$50 for 2kg
3D printer	prusa research	\$1100	N/A	\$1100
LED	eboot/amazon	\$0.076 for 1pc	\$7.6 for 100pc	\$7.6 for 100pc
arduino circuit board (rev3)	arduino	\$27.6 for 1pc	\$27.6 for 1pc	\$27.6 for 1pc
PCB	elegoo/amazon	\$0.375 for 1pc	\$12 for 32pc	\$12 for 32pc
TOTAL		\$6,513.0226	\$437.3	\$6847.3

9.3 Code

```
1  int Upper = 600;
2  bool IgnoreReading = false;
3  int reading = 0;
4  int blinkUp;
5  int blinkDown;
6  int blinkTime;
7
8
9  void setup() {
10     pinMode(13,OUTPUT);
11     pinMode(3,OUTPUT);
12     Serial.begin(9600);
13 }
14
15 void loop() {
16     reading = analogRead(A0);
17     float voltage = (reading);
18     Serial.println(reading);
19
20     //Detecting blinking
21     if (reading >= Upper && IgnoreReading == false) {
22         IgnoreReading = true;
23         blinkUp = millis();
24         digitalWrite(13,HIGH);
25         //Blinking motion for motor
26         analogWrite(3,160);
27         delay(200);
28
29     }
30
31     //Blink over
32     if (IgnoreReading == true) {
33         blinkDown = millis();
34         digitalWrite(13,LOW);
35         IgnoreReading = false;
36         //Blink debounce
37         analogWrite(3,-160);
38         delay(200);
39     }
40 }
41
42
```